



Biodegradable shoes

New Adidas 100% biodegradable kicks are made of Biosteel fiber developed by a German biotech company. <http://dgit.in/adidsbio>



Chat using chemicals

Stanford researchers develop a device that can send instructions using chemicals instead of electric signals. <http://dgit.in/ChmSgnl>

Space Age

was then studied, shining light on the crucial differences between differentiation happening on Earth (at 1g) and in space (almost 0g). It was found that gene markers of new cell formation and regeneration are abnormally expressed in microgravity, with the gene expression for most tissue specific markers not reaching the same levels as they would on Earth. The stem cells didn't differentiate as much or as easily as they would on Earth. Conversely, the gene markers for the processes related with stem cell self-renewal were overexpressed in microgravity, suggesting that the presence of gravity is important for specialisation of the stem cell progenitors! This result conclusively gives a reason for the loss of bone mass experienced by humans who travel into space – our body has a hard time repairing out there!

Another result from the experiment was the observation of stem cells that were flown into space and returned to Earth showed a greater "stemness", which can be described as the ability to self-renew and differentiate into any type of tissue. The space travelling stem cells stay immortal for longer and don't fuss when induced to differentiate into any cell type. The same should happen when the stem cells are inside our body as well, giving them enhanced stemness and giving us a boost of regeneration. Perhaps marketers of commercial spaceflight can use this finding to their advantage!



"I don't mean to show off but my cells have been to outer space!"

Plants vs. Aliens (Advanced Astroculture™)

If we ever hope to colonise Mars, we need to grow plants in space. Moreover, celestial bodies are all really far apart

so space travel takes a long time. Since we haven't developed safe cryogenic hibernation, the ability to grow low mass high energy food aboard the vessel will greatly reduce strains on the resources necessary for inter-planetary exploration and other long distance travel. This project (abbreviated ADVASC) was designed as a nearly autonomous system that required a full-time power supply when it was aboard the ISS, with the crew only having to periodically monitor the plants, take samples at scheduled intervals, and ensure that the hardware was functioning correctly. The ADVASC hardware consisted of a controlled growth chamber that maintained a temperature of 22 degrees Celsius and a relative humidity of 70 percent. The plants were given 18 hour light periods interspersed with six hours of darkness.

On its most recent run, the ADVASC ran three experiments. First, the scientists successfully grew mustard plants (*Arabidopsis thaliana*) from seeds, in space. Though 90 percent of the seeds germinated, only 70 percent of the plants reached maturity after two months. After the success of one seed to seed generation, the next experiment consisted of growing the 'space seeds' from the first generation. While most of the space grown seeds were returned to be planted on Earth (and grew into normal plants with no visible defects), some of them were planted aboard the ISS to grow a back-to-back generation. The third experiment consisted of growing soybeans seed to seed for the first time in space. In all cases, soil and plant tissue samples were also preserved and sent back to Earth for further research.

For the most part, the plants did not grow much differently in space than they did on Earth, when it came to growth rates, number of seeds produced and other physical characteristics. The most notable differences were in branch angles and seed sizes. While branches protrude at an upward angle on Earth, space grown mustard plants had branches that were almost exactly perpendicular to the main stem. Thus it can be inferred that

gravity, and not light, is responsible for the plant branches on Earth that angle towards the sky. With regards to the difference in seed sizes, the soybean space seeds were four percent larger on average, while the mustard space seeds were almost half the size of ground grown seeds. However despite the size differences, they showed similar protein content, germination rates and other physical characteristics compared to Earth grown seeds. By the time you have grandchildren, don't be surprised if your grocery guy is stocking space vegetables!



Space seeds still have upward pointing branches if grown on earth

Fat flames

Knowledge and control of fire is what originally allowed us to transcend the class of animals, ages ago. Ages ahead, in order to not be rendered extinct by fire in space, we need to understand flames and especially how to suppress them. The way fire burns in space is very different from the way it burns on Earth. Here on Earth, gravity causes hotter gases to rise and form the tapered pointy flames that anyone belonging to any civilization will be familiar with. This flow draws in oxygen rich air from the bottom and the flame lives on easily. Moreover, different parts of the flame have different temperatures.

None of this is true for the way fire burns in space. The flame spreads out uniformly into a globe, thanks to the lack of convection due to gravity. The temperature throughout the flame is also uniform, likely due to the same reason. This is far from the complete picture required to properly prevent and extinguish fire in space, whose burning is a complex interaction of fuel vaporisation,